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Hybrid Cooling System for Solar Photovoltaic Panel

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Abstract. Solar photovoltaic (PV) panel is one of the renewable sources of energy and produced daily nowadays. Solar panel systems have efficiency influenced by different factors, such as ambient temperature, solar panel temperature, sunlight, weather, and irradiation. The increasing of the temperature of the solar PV panel decreases its efficiency and lifetime. Thus, to maintain and decrease the temperature of the PV cell, cooling system is required. This paper presents the hybrid (water and air) cooling system method for solar PV panel. The method has been designed and developed to lower and stable the operating temperature of the system considering different weather conditions. The results revealed that the hybrid cooling system has shown improvement of output power solar PV panel as compared with water cooling system only. Furthermore, the proposed method managed to improve efficiency approximately to 4.5%.

1. Introduction

The sources of renewable energy are becoming increasingly common nowadays considering pollution and the sustainability of conventional energy sources. Solar energy is one of the most available and reliable energy today. One of the renewable and clean energy is the use of photovoltaic (PV) systems that convert electrical energy by sunlight. This is the type of sustainable energy source, free from pollution, energy technology which reduces global warming. At the same time, PV technologies lowers billing operating expenses and minimum operating costs. The advantages of solar photovoltaic (PV) technology are demonstrated by the highest maintenance and output power as compared with other renewable and clean energy technologies [1].

There are advantages using PV technology, but solar PV system has common problems, for examples, dust, hail, and surface operating temperature, which can affect the efficiency of the solar energy conversion system [2]. Exogenous climate parameters such as ambient temperature, relative humidity, wind speed, accumulated dust, and solar radiation are the common natural factors influencing the surface temperature of the PV module [3]. According to the authors in [4], increasing the surface temperature of 1°C PV module may decrease efficiency approximately to 0.5%. Therefore, the increase in temperature causes not all solar energy to be absorbed by PV cells to convert into electrical energy. To meet the electrical energy-saving catalyst, the remaining solar energy is converted into heat. In addition, this wasted heat will reduce the overall conversion efficiency [5].

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Solar cooling technology has been the subject of numerous studies to conserve solar energy resources. Solar PV conversion cooling technology utilizes photovoltaic cells to convert solar energy to electrical energy, which is then utilized to induce vapour compression refrigeration [6]. Solar absorption cooling is the most cost-effective method of solar photovoltaic conversion cooling. Solar collectors are acclimated to convert solar energy to thermal energy, which is then applied to induce absorption cooling in solar air-cooling technology [7]. Economic performance is a critical factor in solar cooling technology application. Numerous researchers have conducted economic analyses of solar cooling systems [8-11].

The aim of this research is to examine the influence of different weather conditions on the output performance of solar panels. Furthermore, the performance of solar panels can be influenced by various factors, including ambient temperature, solar panel temperature, weather, sunlight, and solar radiation. Thus, hybrid cooling system consisting of water- and air-cooling systems have been designed to overcome this problem in terms of solar PV power output.

2. Methodology

The research intends to improve the performance of the cooling system for solar PV systems, by introducing hybrid cooling system that consists of water- and air-cooling system. **Figure 1** shows the design flow for a PV system with a hybrid cooling system and work process. A hybrid cooling system using temperature sensor, Arduino, blower fan and water pump is designed and built. The Arduino is programmed to trigger the water pump when the temperature of the PV panel is more than 50 °C and trigger blower fan when the temperature is more than 50 °C.

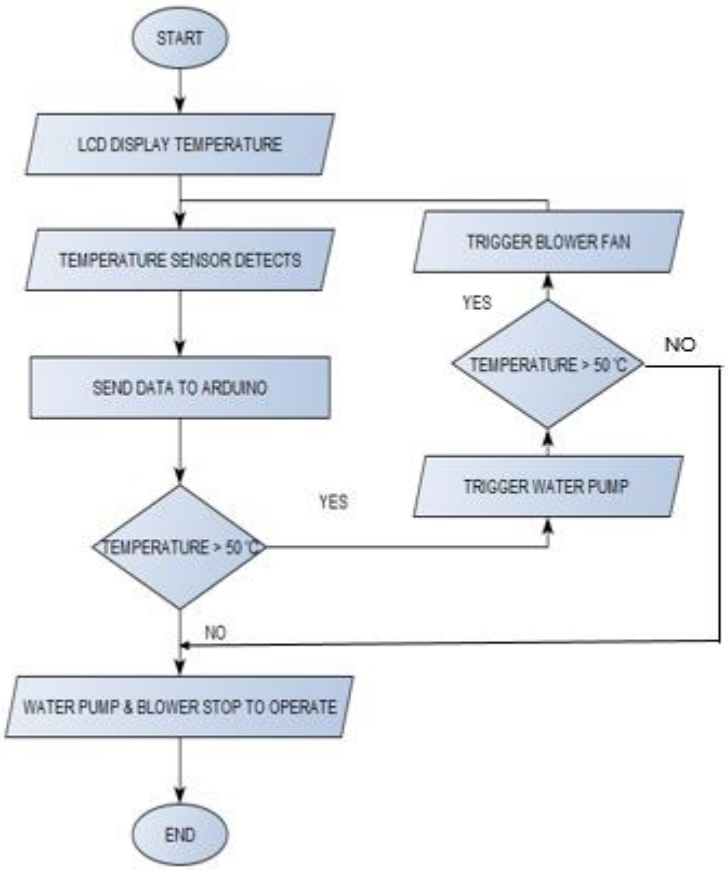


Figure 1. Flowchart of the Hybrid Cooling System

The research is divided into two stages: the software implementation using Proteus 8 and hardware design of hybrid cooling system. **Figure 2** shows the design for controlling the water pump system while **Figure 3** shows air exhaust blower system connection.

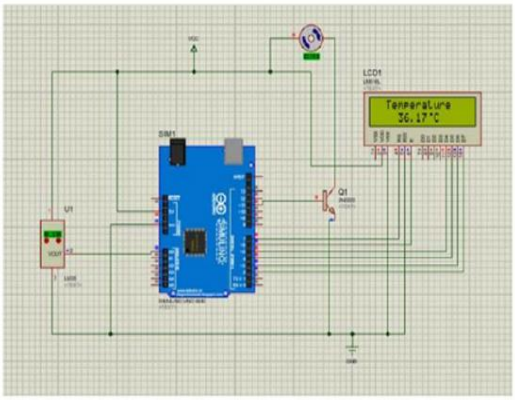


Figure 2. Circuit connection to control water pump system

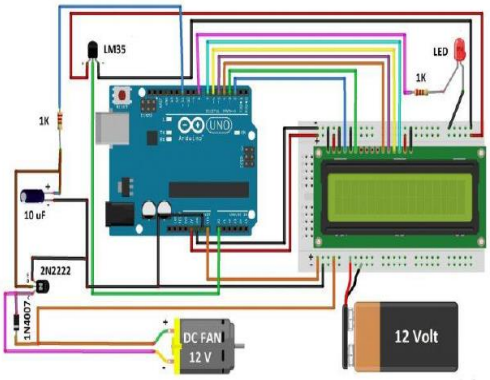


Figure 3. Circuit connection to control exhaust blower system

For conducting the experiment, the setup of the hybrid cooling system is depicted as in **Figure 4.**

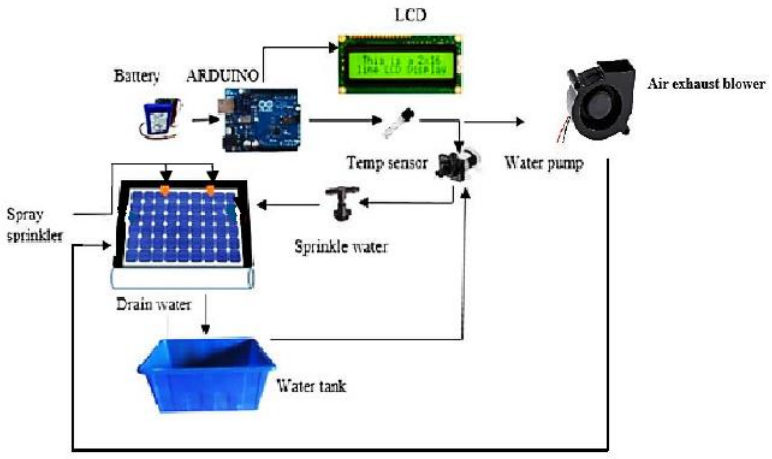


Figure 4. The setup of the water- and air-cooling PV system

In this experiment, a 50 W solar panel is used. At the top of solar PV, two spray sprinklers are used, while at the bottom, two air exhaust blowers are used to draw excess heat away from the solar panel, thereby indirectly reducing heat on the solar panel. The temperature sensor is attached to the solar panel's top surface to detect the heat generated by the panel's surface. The temperature value detected by the temperature sensor is used by the Arduino to power on the water pump and the water exhaust blower. The temperature reading value and the amount of energy produced can be viewed on the Liquid Crystal Display (LCD). Additionally, a 12 V battery is used in this experiment to power the Arduino, LCD, water pump, and water exhaust blower. Water tanks used to store and recycle the water sprayed by this spray sprinkler can help to reduce the rate of water consumption in this experiment indirectly. This hybrid cooling system consists of nine parts which are listed and tabulated in **Table 1**.

Table 1. Components of the cooling system

Item	Description	Unit
PV Solar Module	V_{smax} = 1000 V, P_{max} = 50 W, V_{oc} = 21.6 A, I_{mp} = 2.77 A, V_{mp} = 18 V, I_{sc} = 2.9 A, Irradiance = 1000 W/m^2, Cell Temperature = -40°C ~ +80°C	1
Water Pump	DC motor with high speed, 5 V and 83.5 V	1
Sprinkler Spray	For cleaning solar PV panel	2
Water Tank	Plastic material	1
Air Exhaust Blower Fan	High speed around 1000 rpm with 12 V	2
Arduino	Input voltage 9V and the output voltage are 5 V and 3.3 V	1
Temperature Sensor	Negative Temperature Coefficient (NTC) temperature probe (10k 3950)	1
Battery	12 V	1
LCD Display	$V_{operating}$ = 4.7 V to 5.3 V, $I_{consumption}$ = 1 mA	1

In this research, the energy source which is a Direct Current (DC) battery is used for turning on the air exhaust blower fan and water pump. Water is sprayed with two sprinkler which is installed on the front top of the solar PV panel. Then, the water will flow to the atmosphere below the solar PV panel from an open drain that releases warmth and flow to the water tank. Next, the exhaust blower fan will automatically turn on and inhales the excess heat contained on the solar PV panel. Throughout the solar PV panel, the heat absorbed through this air exhaust blower and the heat is released.

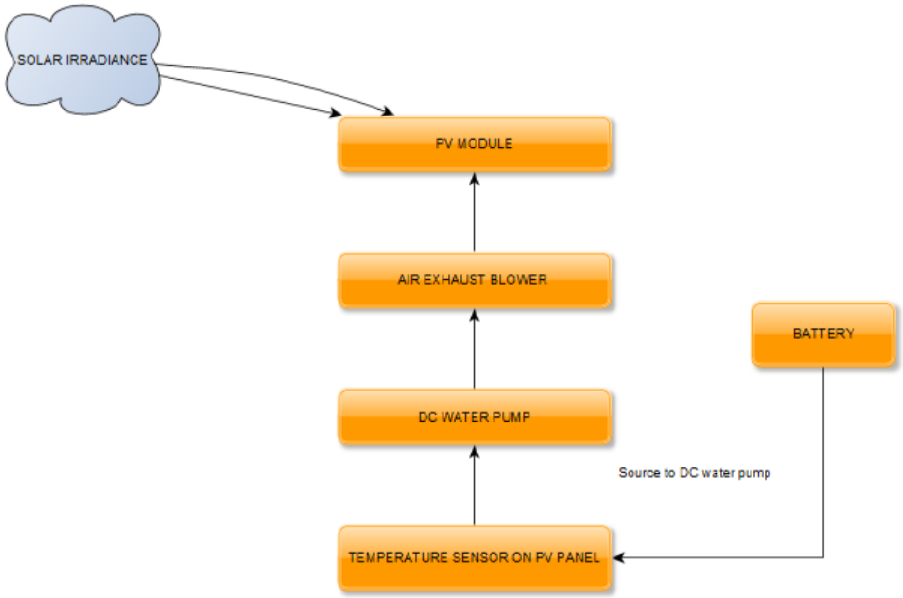


Figure 5. Block diagram of the hybrid cooling system

To understand the concept of the hybrid cooling system, **Figure 5** provides an overview of the hybrid cooling systems. As can be seen, solar panels require light and irradiance to generate output energy. The battery powers the temperature sensor on the PV panel. When the temperature rises above a certain point, the water pump and exhaust water blower are activated to cool the PV panel.

To maximize the PV panel's output power, the calculation of ideal load is very essential. Although the maximum power, P_{max} and current maximum power, I_{mp} are specified, both parameters do not include the load resistance, $R_{load.}$ The resistor used must be approximately equal to the P_{max} and I_{mp}. If the values of P_{max} and I_{mp} are unknown, the open-circuit voltage, V_{oc} and short circuit current, I_{sc} provide reasonable approximations. Typically, the module's back will have a label listing P_{max} and I_{mp}. A digital multimeter can be used to measure V_{oc} and I_{sc}. The module generates hundreds of watts of power, necessitating the use of a high-wattage power resistor. However, because the measurements are for a brief period, a power resistor at full power is not required. Equation (1) below shows the calculation of resistor given that the power is 50 W and current maximum power flow is 2.77 A.

$$P = I^2 R \tag{1}$$

$$R = \frac{P}{I^2} = \frac{50\ W}{(2.77A)^2} = 6.51\ \Omega$$

From this equation, the initial load for solar panels 50 W to use is 6.51 Ω.

To improve the performance of solar PV panel, the efficiency calculation is very essential. The efficiency formula for solar PV panel is expressed as in Equation (2):

$$\eta = \frac{P_{max}}{Area\ x\ 1000\ W/m^2}\ x\ 100\% \tag{2}$$

Where P_{max} = maximum panel power (W)
At STC, solar irradiance = 1000 W/m^2
Area panel = (0.70 x 0.54) m^2

3. Solar Photovoltaic Panel with Hybrid Cooling System

The load connected to the PV panel is shown in this section which are PV panel, digital multimeter and ceramic tube adjustable rheostat resistor. **Figure 6** demonstrates the connection of solar PV panel to the load.



Figure 6. Connection of solar PV panel to the load

Figure 7(a) and **Figure 7(b)** displays the water-cooling system at the front of PV panel and air-cooling system at the back of PV panel respectively. Digital multimeter has been used for recording the voltage and current value. The load used in this experiment is a rheostat. The LCD displays the temperature measured by the NTC temperature probe (10k 3950) sensor where had been installed in the front of the PV panel to control or turn off the water-cooling system.

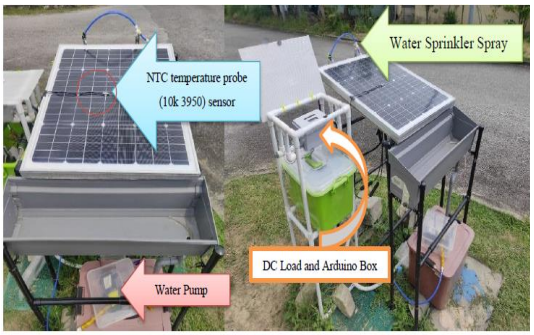


Figure 7(a). Water Cooling System



Figure 7(b). Air Cooling System

Figure 7(b) shows air cooling system were put together by combining it with water cooling in the PV conversion system. The air-cooled PV panel is applied to absorb the heat produced by finding high-energy photon loss and low-energy photon transmission losses from the PV panel surface area. At the front top of the PV panel, the water-cooling module is positioned. The temperature has increased significantly as the photovoltaic device is exposed to solar radiation. There is a temperature differential between the water cooler and the air temperature sensor due to the dispersion of the charge carrier in the PV panel if the top and bottom surfaces change in temperature. The energy obtained from the solar cells is dispersed and deposited in resistance. Therefore, heat sink is used to cool the surface of the PV panel's heat dissipation.

4. Results and Discussion

Hybrid cooling system was designed and built to improve the performance of solar PV panel in terms of output power and efficiency. This analysis is conducted by measurement of the voltage, current and power against time. Rheostat Ceramic Tube Adjustable Resistor was used as the load on the PV panel. This experiment utilized a load of 100 W and a resistance of 100 Ω. The load was adjusted starting from 0 Ω to 100 Ω in increments of 0 Ω.

This experiment was conducted during the peak of the PV panel's output voltage to determine the optimal number of resistors to be used. The load was adjusted to the 5 Ω interval resistor value, and all data were collected and tabulated in **Table 2.** The output power was measured simultaneously with each resistor value change. The higher output power is 34.28 W at a resistor value of 6.5 Ω, according to the data collected and highlighted in **Table 2**. The resistor value should be selected to maximize the output power while the load voltage increases in lockstep with the resistance and the current is decreased slightly. The maximum output power and resistor values in this result remain variable to allow for testing of the reference PV panel with hybrid cooling system and water-cooling system.

Table 2. Resistor Value

Resistance, Ω	Voltage,V	Current,I	Power,W
0	0	2.79	0.00
2	10.2	3.12	31.82
3.7	10.34	3.1	32.05
4.5	12.63	2.53	31.95
5.3	12.88	2.49	32.07
6.5	14.65	2.34	34.28
7.4	12.35	1.62	20.01
10.7	15.74	1.34	21.09
15.8	17.51	1.06	18.56
20.6	15.46	0.75	11.60
25.4	15.96	0.61	9.74
29.7	17.68	0.51	9.02
30.6	18.29	0.57	10.43
35.6	18.22	0.47	8.56
40.3	18.44	0.4	7.38
45.5	18.58	0.35	6.50
50.5	18.73	0.32	5.99
55.5	18.86	0.29	5.47
60.7	18.99	0.26	4.94
65.9	19.08	0.24	4.58
70	19.03	0.22	4.19
75.5	19.3	0.21	4.05
80.5	19.05	0.19	3.62
85.8	19.29	0.19	3.67
90.2	19.18	0.18	3.45
95.7	19.32	0.16	3.09
100	19.17	0.15	2.88

This experiment was conducted at the Police District Headquarters (IPD) Barat Daya, Penang, where the heat was very high and was carried out starting from 8:00 am until 5:00 pm. PV panels with a maximum value of 50 W and 18 V were used in this experiment. PV panels are installed in the field facing the rising sun, which is to the east at a 30° tilt angle. The results shown below is for a resistor with a fixed value of 6.5 Ω. The resistor set to a fixed value on the PV panel is used to measure the voltage and current values of the PV panel with a hybrid cooling system and water-cooling system. After recording all the data, the output power is calculated.

Table 3(a). Solar PV panel water cooling system

Water Cooling System				
Time	Temperature (^{0}C)	Voltage (V)	Current (A)	Power (W)
8.00 am	24.7	3.33	0.11	0.37
8.30 am	27.4	4.8	0.18	0.86
9.00 am	33.9	6.97	0.26	1.81
9.30 am	38.2	8.12	0.32	2.60
10.00 am	34.9	10.23	0.45	4.60
10.30 am	29.4	10.38	0.25	2.60
11.00 am	31.2	11.46	1.34	15.36
11.30 am	31.3	11.47	1.33	15.26
12.00 pm	53.7	17.52	2.10	36.79
12.30 pm	52.3	17.61	2.14	37.69
1.00 pm	52.1	16.87	1.99	33.57
1.30 pm	51.3	16.33	1.9	31.03
2.00 pm	50.1	14.87	0.54	8.03
2.30 pm	50.4	14.55	0.51	7.42
3.00 pm	50.3	14.21	0.49	6.96
3.30 pm	50.2	14.65	0.47	6.89
4.00 pm	38.4	12.38	0.41	5.08
4.30 pm	38.9	11.26	0.36	4.05
5.00 pm	37.3	10.25	0.59	6.05

A comparative study has been made and tabulated as in **Table 3(a)** and **Table 3(b).** The result shows that PV panel with hybrid cooling systems increase the output power significantly in the range 69% until 83% from 10 am until 4.30 pm as highlighted in the table. In terms of temperature, temperature of the PV panel with water cooling system increased slightly from 24.7 °C at 8 am to 27.4 °C at 8.30 am, while the temperature of the hybrid cooling system increased slightly from 25.7 °C at 8 am to 27.4 °C at 8.30 am.

Table 3(b). Solar PV panel hybrid cooling system

Hybrid (Water and Air) Cooling System				
Time	Temperature (^{0}C)	Voltage (V)	Current (A)	Power (W)
8.00 am	25.7	3.45	0.12	0.41
8.30 am	27.4	4.8	0.18	0.86
9.00 am	35.9	7.21	0.28	2.02
9.30 am	38.2	8.11	0.32	2.60
10.00 am	34.9	10.23	1.45	14.83
10.30 am	29.4	10.38	1.25	12.98
11.00 am	42.4	18.48	2.61	48.23
11.30 am	45.8	17.69	2.57	45.46
12.00 pm	46.3	16.98	2.13	36.17
12.30 pm	47.3	17.61	2.14	37.69
1.00 pm	50.3	17.79	2.11	37.54
1.30 pm	51.1	17.43	2.05	35.73
2.00 pm	42.9	18.05	2.15	38.81
2.30 pm	43.4	18.25	2.1	38.33
3.00 pm	46.9	15.67	1.78	27.89
3.30 pm	35.7	8.31	0.97	8.06
4.00 pm	38.4	17.69	2.04	36.09
4.30 pm	38.9	17.45	1.31	22.86
5.00 pm	34.8	7.4	0.81	5.99

According to **Figure 8**, after 9.30 am, the temperature of the PV panel decreases slightly until 10.30 am for water cooling, and there is a significant difference between water cooling system and hybrid cooling system as the temperature increases. The blue line represents a PV panel that is only cooled by water and the orange line represents a PV panel that is cooled by both water and air. Since the PV panel utilizes water cooling system, the temperature on the PV panel increases rapidly, which influences the PV panel's output current. The temperature on the PV panel with water cooling system continued to rise until it reached 53.7 °C at 12 pm when it reached its maximum output value and then decreased gradually until 5 pm. As can be seen, by utilizing hybrid cooling systems, PV panels can maintain a constant temperature and even reduce heat generated by the PV panel indirectly. This can also help maintain the PV panel's output.

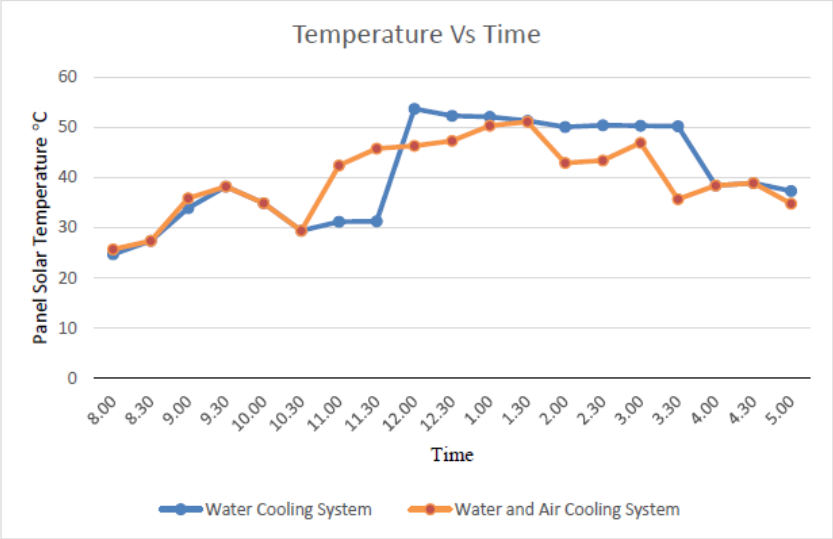


Figure 8. Comparison of Water-Cooling System and Hybrid Cooling System for Temperature against Time

Figure 9 represents the output voltage on the PV panel for both systems. Once analysed, the output voltage on the PV panel did not change significantly because the voltage was unrelated to the PV panel's temperature. The temperature with water cooling and with hybrid cooling system is identical at the start, but there is a difference after 10.30 am due to the peak sun time. However, by utilizing a hybrid cooling system on the PV panel, the output voltage can be maintained at a higher level as compared with water cooling system. The graph demonstrates that the PV panel does not have a significant effect on the output voltage as the temperature of the panel increases because the higher the temperature of the panel, the stronger the radiation emitted by the panel.

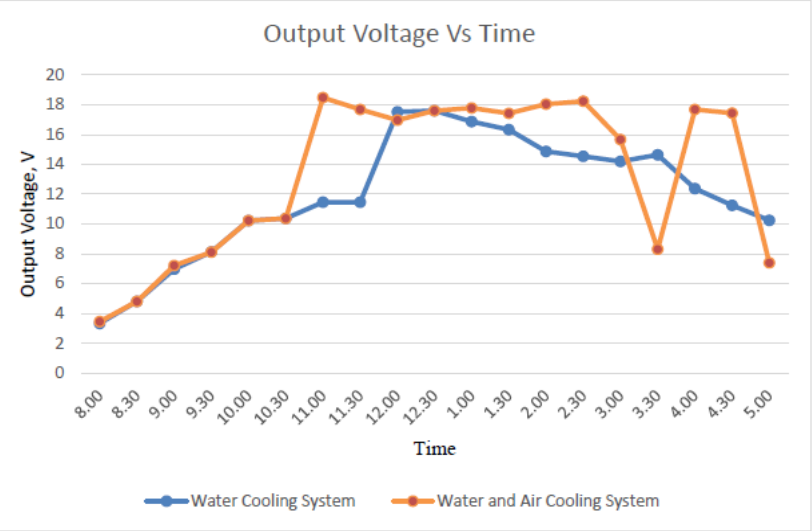


Figure 9. Comparison of Water-Cooling System and Hybrid Cooling System for Output Voltage against Time

The output current readings analysed during the experiments are shown in **Figure 10**. A digital multimeter was used to measure the current output in series mode. According to the obtained values, the PV panel with water cooling system has a lower average output current than the PV panel with a hybrid cooling system. At 8:00 am, the initial output current for the water-cooling system was at 0.12 A with 0.11 % efficiency, which is consistent with the values obtained throughout the experiments for the hybrid cooling system's PV panel.

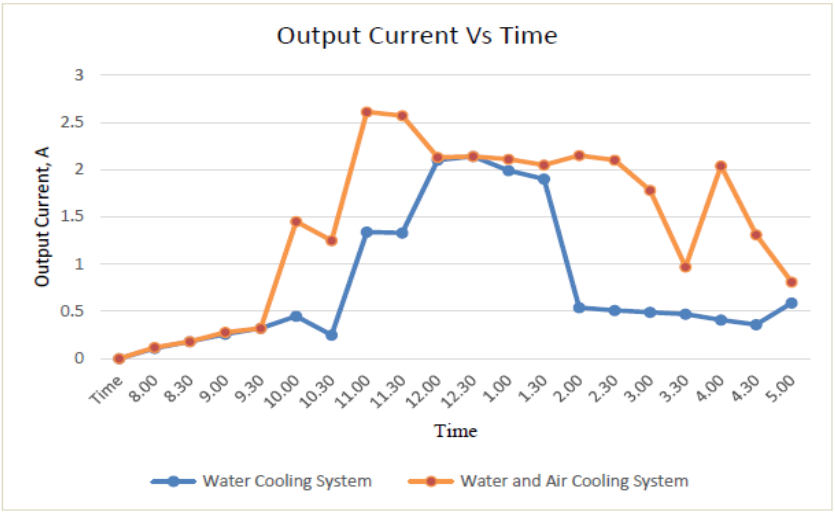


Figure 10. Comparison of Water-Cooling System and Hybrid Cooling System for Output Current against Time

Figure 11 illustrates the output power performance of both systems. As can be seen, PV panels with water cooling systems produce slightly less energy efficiency of 10 % but hybrid cooling system produce 13 % efficiency. The output power is almost the same at first, between 0.10 % to 4 % efficiency, but after 10.30 am, when the PV panel only uses water, there is a loss of power as the temperature rises, affecting the output power by 8 % efficiency.

At 12.00 noon, the maximum value of output power water cooling system is only 36.79 W. After that, the power output is unsustainable, and the efficiency slightly decreases at 9 %. Due to the high temperature around 12.00 noon, the voltage value drops suddenly by 8 % efficiency, affecting the output of the PV panel. As illustrated in the figure, the output power for hybrid cooling system can last longer than the PV panel with water cooling system with 13 % efficiency differences. This indicates that when hybrid cooling systems are used, the output power of PV panel is more efficient and can withstand higher loads for longer periods of time.

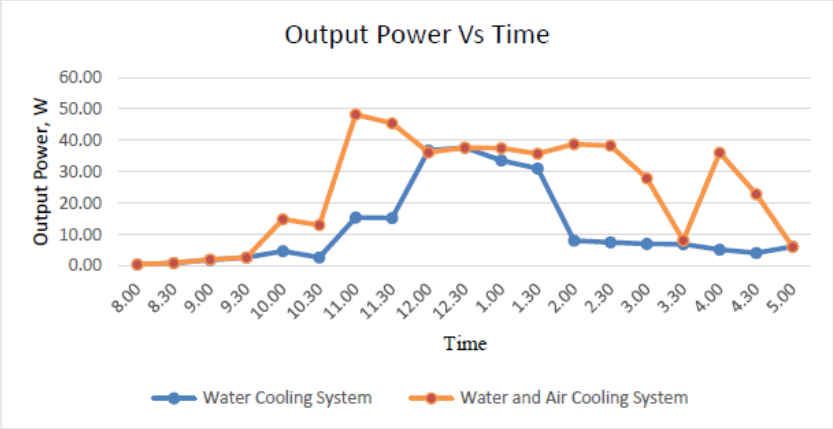


Figure 11. Comparison of Water-Cooling System and Hybrid Cooling System for Output Current against Time

Table 4 tabulates the average temperature, voltage, current and power readings of Solar PV panel with water cooling system and hybrid cooling system for 15 days.

Table 4. Average value and Standard Deviation value of Temperature, Current and Power for Solar PV panel with water cooling and hybrid cooling system

Time	Temperature (^{0}C)		Voltage (V)		Current (A)		Power (W)	
	Water Cooling	Hybrid Cooling	Water Cooling	Hybrid Cooling	Water Cooling	Hybrid Cooling	Water Cooling	Hybrid Cooling
8.00 am	27.91	27.91	2.83	3.21	0.10	0.17	0.28	0.68
8.30 am	30.89	30.69	4.54	4.76	0.16	0.24	0.76	1.23
9.00 am	34.97	34.69	5.83	6.86	0.20	0.29	1.26	2.60
9.30 am	38.46	37.07	8.05	8.97	0.27	0.45	2.27	4.51
10.00 am	41.83	40.53	10.16	10.64	0.54	0.73	5.85	8.41
10.30 am	44.80	43.03	11.48	11.36	0.93	0.94	11.11	11.21
11.00 am	47.95	45.28	13.23	12.71	1.29	1.32	17.93	18.17
11.30 am	49.14	47.05	14.59	14.13	1.54	1.57	22.59	23.58

Time	Temperature (^{0}C)		Voltage (V)		Current (A)		Power (W)	
	Water Cooling	Hybrid Cooling	Water Cooling	Hybrid Cooling	Water Cooling	Hybrid Cooling	Water Cooling	Hybrid Cooling
12.00 pm	51.07	47.72	15.30	15.40	1.68	1.66	27.16	26.43
12.30 pm	52.41	47.97	16.12	16.01	1.83	1.61	30.67	26.50
1.00 pm	53.10	46.63	16.70	16.87	1.97	1.79	33.80	31.20
1.30 pm	53.11	47.97	16.71	16.80	1.93	1.79	32.60	30.83
2.00 pm	53.00	44.77	16.39	15.72	1.90	1.72	31.27	27.89
2.30 pm	52.54	44.59	15.88	15.22	1.76	1.75	27.87	27.73
3.00 pm	51.45	46.48	15.03	14.67	1.75	1.65	26.33	25.68
3.30 pm	50.74	45.41	14.42	14.51	1.57	1.62	22.58	25.37
4.00 pm	48.23	44.99	13.32	17.02	1.25	1.97	**17.29**	**34.48**
4.30 pm	47.08	43.17	12.27	17.36	1.20	1.87	15.47	32.22
5.00 pm	45.45	41.79	10.78	10.89	0.85	1.26	9.85	15.52
Standard Deviation	7.5558	5.7149	4.1576	4.1534	0.6452	0.5961	11.4665	11.1986

Based on **Table 4,** it can be noted that the temperature for solar PV panel with hybrid cooling system reduces to 0.2 °C differences at 8.30 am. At 2.00 pm, the temperature of hybrid cooling system reduces to 44.77°C from 53°C which 8.23 °C differences as compared with water cooling system. It implies that hybrid cooling system can reduces the temperature of solar PV panel. As highlighted in **Table 4**, the output power increase at 8.00 am until 11.30 am in the range of 0.1 W to 2.56 W. While at 3.30 pm to 5.00 pm, the output power increases in the range of 2.79 W to 17.19 W. However, the average output power reduces for hybrid cooling system at 12.00 pm until 3.00 pm as highlighted in the table. This is due to cloudy weather condition, dust and dirt shading that creates barrier in the path of incoming sunlight which reduces power output from 0.14 W to 4.1 W and performance of the panel. In terms of standard deviation, it can be noted that the value of temperature, voltage, current and power for hybrid cooling system are better compared to water cooling system. As highlighted in **Table 4**, this indicate the accuracy of the average data where the lower the standard deviation, the closer the average data tend to be the expected value.

The efficiency of solar PV panel with hybrid cooling system and water-cooling system are calculated. For example, based on **Table 4**, the average P_{max} of the panel is 34.48 W.

$$\eta = \frac{P_{max}}{Area\ x\ 1000\frac{W}{m^2}}\ x\ 100\% = \frac{34.48\ W}{(0.7\ x\ 0.54)m^2\ x\ 1000\frac{W}{m^2}}\ x\ 100\% = 9.12\%$$

$$\eta = \frac{P_{max}}{Area\ x\ 1000\frac{W}{m^2}}\ x\ 100\% = \frac{17.29\ W}{(0.7\ x\ 0.54)m^2\ x\ 1000\frac{W}{m^2}}\ x\ 100\% = 4.57\%$$

From the value obtained, the efficiency of solar PV panel is increased from 4.57 % with water cooling system to 9.12 % with hybrid cooling system. This proves that the designed hybrid cooling system increases the efficiency of solar panel.

In terms of output power produced against days, it can be noted that the average output power data for 15 days with hybrid cooling system is significantly improved. This is shown in **Figure 12**. Furthermore, water cooling system has lower output power as compared with hybrid cooling system as sunlight produces strong radiation at high temperatures for PV panel. When the amount of sunlight hitting the photovoltaic panel increases, this hybrid cooling system can lower the temperature of the PV panel and maintain the maximum output power as well as energy generated in the long run. It can be noted that it also produces higher voltages and currents, increased output power and efficiency by 14 % on the eleventh day. Nevertheless, there was a slightly decrease on the ninth, and twelfth day with efficiencies of 9% and 10% due to adverse weather conditions and inadequate sunlight lighting as well as hardware limitations, which prevented the PV panel from reaching its maximum output rate.

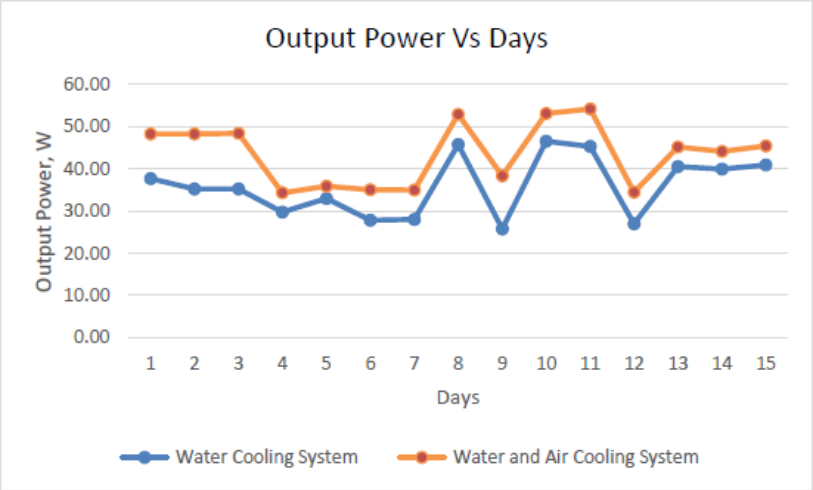


Figure 12. Comparison of Water-Cooling System and Hybrid Cooling System for Output Power against 15 Days

5. Conclusion

Hybrid cooling system is designed for solar panel that will increase the PV panel performance. Thus, by using this hybrid (water and air) cooling system, the maximum output energy produced by photovoltaics is maintained in the long run for the whole 15 days compared to water cooling system. In addition, by using this system, dirt, or dust on the surface of the solar panel can be removed, thus maximizing the amount of power and energy generated by solar PV panel. It can be concluded that this system may help extend the lifetime of a solar panel by preventing it from being damaged or overheating.

References

[1] Akbarzadeh A, Wadowski T. Heat pipe-based cooling systems for photovoltaic cells under concentrated solar radiation. 1996 *J. Appl. Ther. Eng.* **16** 81–87.

[2] Rodrigues E., Melicio R, Mendes V.M.F. and Catala˜o J.P.S., Simulation of a solar cell considering single-diode equivalent circuit model. 2011 *Proc. Int. Conf. on Renewable Energies and Power Quality*, Spain.

[3] Siecker J., Kusakana K. and Numbi B.P., A review of solar photovoltaic systems cooling technologies. 2017 *J. Ren. and Sust. Ene. Rev.* **79** 192-203.

[4] Chaniotakis E. Modelling and analysis of water-cooled photovoltaics, M.Sc. thesis, Faculty of Energy System and Environment, Department of Mechanical Engineering, University of Strathclyde, Glasgow, Scotland; 2001.

[5] Dubey S and Tiwari G.N. Thermal modeling of a combined system of photovoltaic thermal (PV/T) solar water heater. 2008 *J. Sol. Energy* **82** 602–612.

[6] Batoul H. *Flow Simulation Improves Photovoltaic Solar Panel Performance*, Technical Report, Schueco International, Paris, France, September 2008.

[7] Tonui J.K. and Tripanagnostopoulos Y. Improved PV/T solar collectors with heat extraction by forced or natural air circulation. 2007 *J. Renew. Energy* **32** 623–637.

[8] Moharram K.A., Abd-Elhady M.S., Kandil H.A. and El-Sherif H., Enhancing the performance of photovoltaic panels by water cooling. 2013 *J. Ain Shams Eng*. **4** 869-877.

[9] Gratela R.O., Martes J.A.S., Pagatpatan G.I., Fernandez E.O., Amado T.M. and Valenzuela I.C. Hybrid Cooling System for Solar PV Panel. 2020 *Int. J. of Eme. Tre. In Eng. Resear*. **8** 4079-4083.

[10] Shah J., Chowdhary A., Patel D., Thakker D., Talaviya U. and Shah P. Cooling Techniques of Solar Photovoltaic Panels: A Critical Review. 2022 *Int. J. of Eng. Res. & Tech.* **11** 449-460.

[11] Homadi, A., Hall, T. and Whitman L. Study a novel hybrid system for cooling solar panels and generate power. 2020 *J. Appl. Ther. Eng.* **179** 1–8.

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